ECE2002
DIGITAL LOGIC DESIGN

Module 3 – Lecture 1

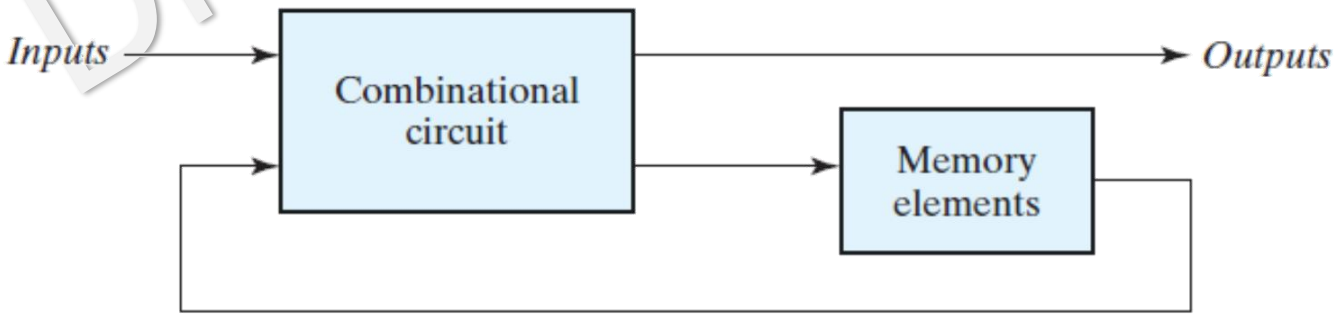
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Sequential Circuits

- Almost all electronic consumer products have the ability to send, receive, store, retrieve, and process information represented in a binary format.
- The technology enabling and supporting these devices is critically dependent on electronic components that can store information, i.e., have memory.
- Sequential circuits consist of a combinational circuit to which storage elements are connected to form a feedback path.
- The storage elements are devices capable of storing binary information.
- The binary information stored in these elements at any given time defines the state of the sequential circuit at that time.

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Sequential Circuits

- There are two main types of sequential circuits, and their classification is a function of the timing of their signals:
    - *Synchronous sequential circuit:* behaviour is defined from the knowledge of its signals at discrete instants of time.
    - *Asynchronous sequential circuit:* behaviour depends on the input signals at any instant of time and the order in which the inputs change.
- Synchronization is achieved by a timing device called a clock generator, which provides a clock signal having the form of a periodic train of clock pulses.
- For example, a circuit that is to add and store two binary numbers would compute their sum from the values of the numbers and store the sum at the occurrence of a clock pulse.

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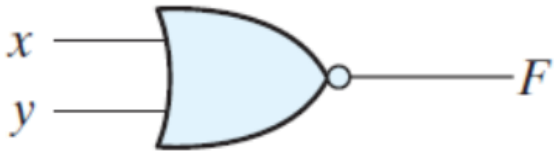
Latches

- Asynchronous sequential circuit that can store 1-bit information
- Storage elements that operate with signal levels (rather than signal transitions)
- There are two stable states: **set** and **reset**
- The set state is denoted by 1, the reset state is represented by 0
- Has feedback mechanism allowing them to maintain the current state until next input is applied
- Level sensitive devices
- Applications
  - used to keep the conditions of the bits to encode binary numbers
  - computing as well as data storage
  - used in power gating and clock as a storage device

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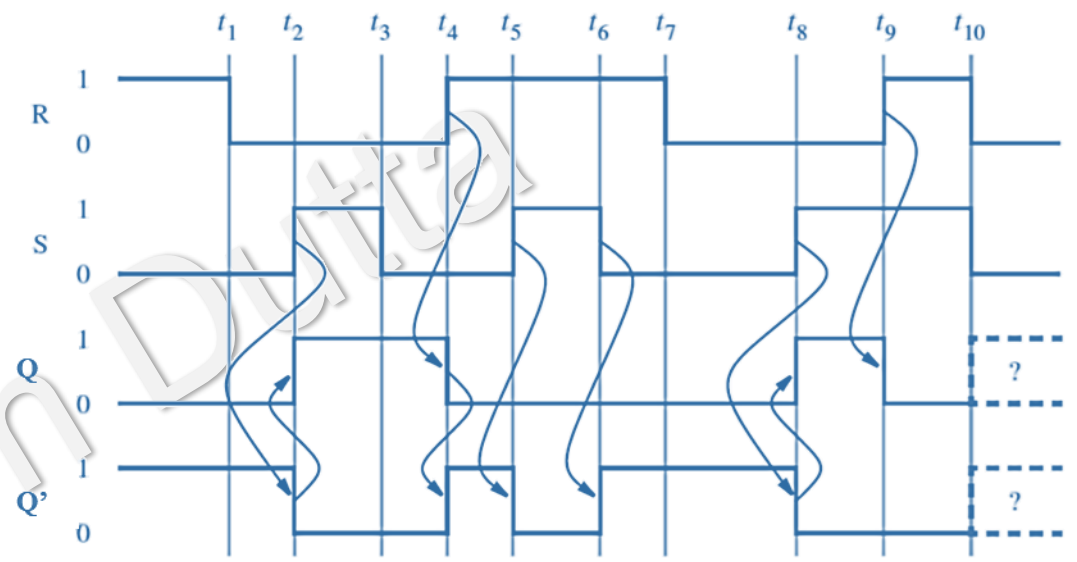
Latches

SR (Set-Reset) Latch with NOR Gates

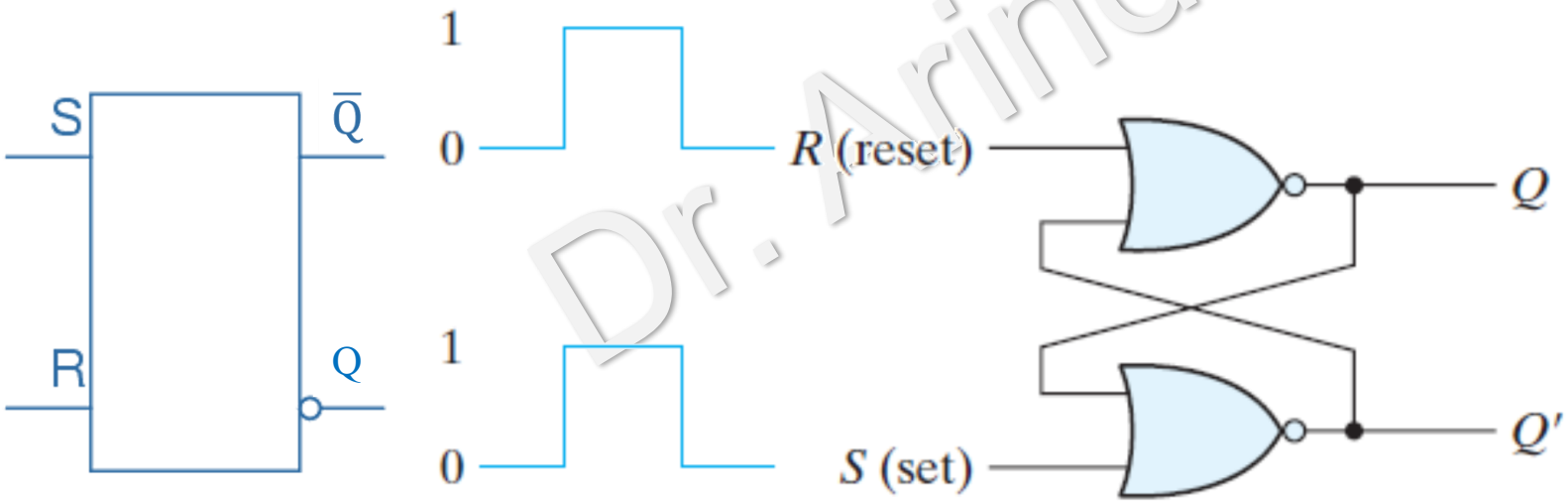


x	y	F
0	0	1
0	1	0
1	0	0
1	1	0

If one input is high (1), output is low (0)



Timing diagram of NOR SR latch

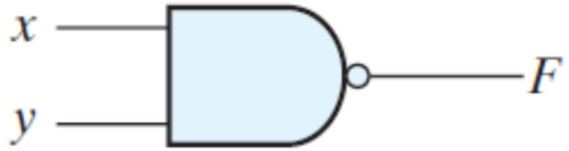


S	R	Q	$\bar{Q}$	State
0	0	Q	$\bar{Q}$	Hold/Unchanged
0	1	0	1	Reset
1	0	1	0	Set
1	1	x	x	Invalid

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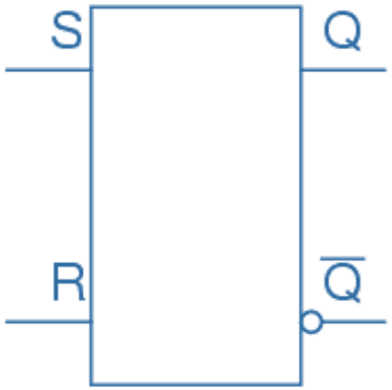
Latches

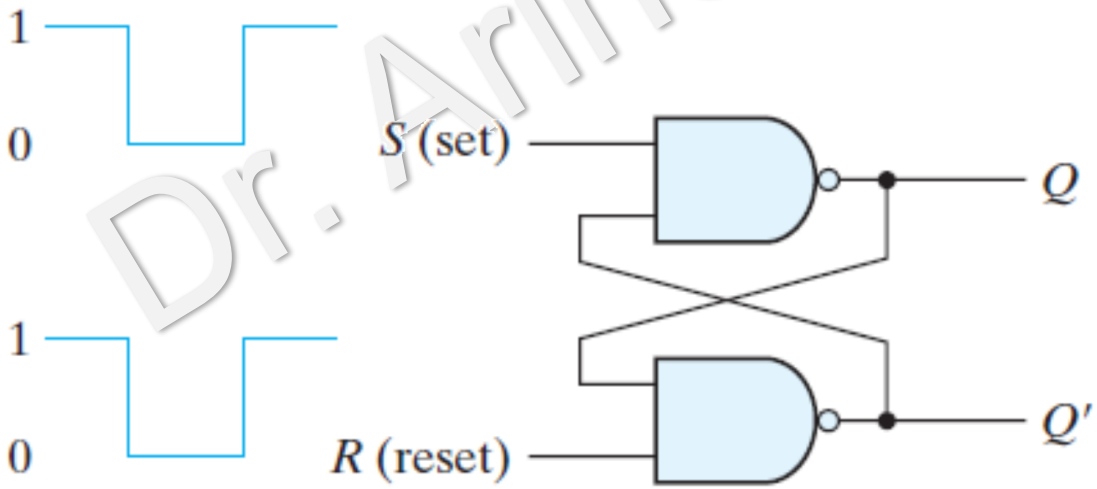
SR Latch with NAND Gates



x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

If one input is low (0), output is high (1)





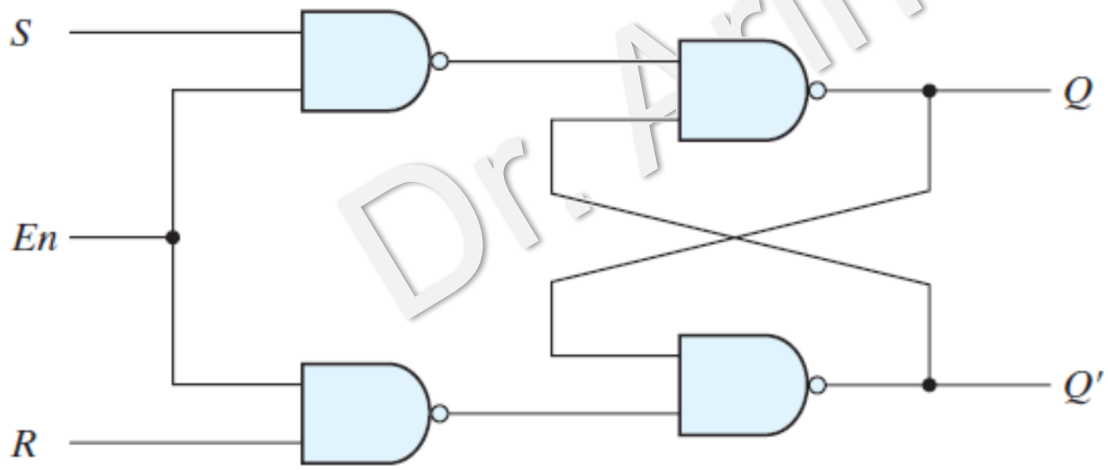
S	R	Q	Q̄	State
0	0	x	x	Invalid
0	1	1	0	Set
1	0	0	1	Reset
1	1	Q	Q̄	Hold/Unchanged

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Latches

SR Latch with Control Input

- It consists of the basic SR latch and two additional NAND gates.
- The control input En acts as an enable signal for the other two inputs.
- The outputs of the NAND gates stay at the logic-1 level as long as the enable signal remains at 0.
- When the enable input goes to 1, information from the S or R input is allowed to affect the latch.



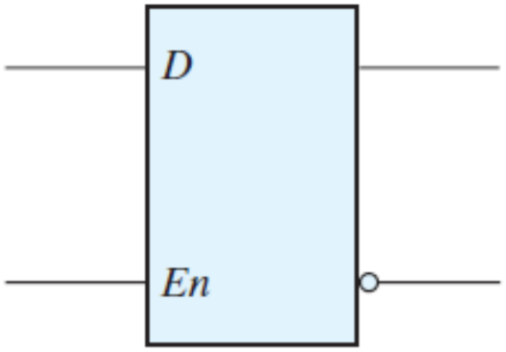
En	S	R	Q_{n+1}	State
0	x	x	Q_n	Hold
1	0	0	Q_n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	x	Indeterminate

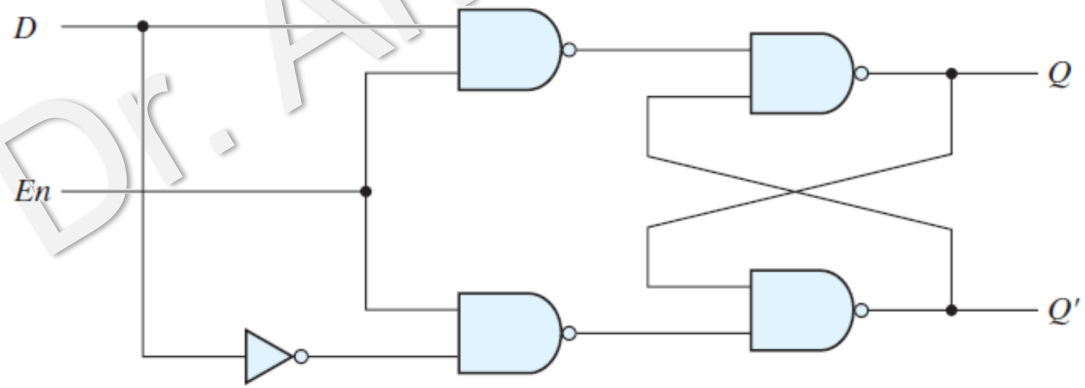
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Latches

D Latch (Transparent Latch)

- D latch eliminates the undesirable condition of the indeterminate state in the SR latch.
- It ensures that inputs S and R are never equal to 1 at the same time.
- The D latch receives that designation from its ability to hold *data* in its internal storage.
- It is suited for use as a temporary storage for binary information between a unit and its environment.





En	D	Q_{n+1}	State
0	x	Q_n	Hold
1	0	0	Reset
1	1	1	Set

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Flip-Flops

- Synchronous or asynchronous sequential digital circuit having two stable states; used to store one bit of binary data
- Fundamental building blocks of all memory devices
- The clock input is used to trigger the flip-flop so that it can change states of its outputs
- The state of a latch or flip-flop is switched by a change in the control input. This momentary change is called a trigger, and the transition it causes is said to trigger the flip-flop.
- The problem with latches is that it responds to a change in the level of a clock pulse (Fig. a).
- The key to the proper operation of a flip-flop is to trigger it only during a signal transition (Fig. b and c).

Applications: frequency dividers, counters, storage registers, shift registers, data storage, data transfer

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(a) Response to positive level



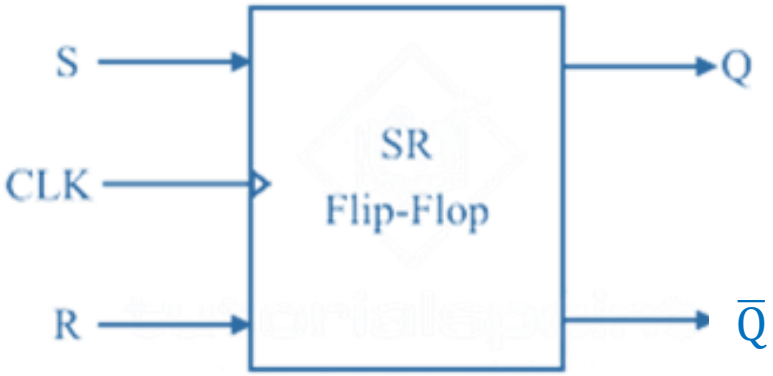
(c) Negative-edge response

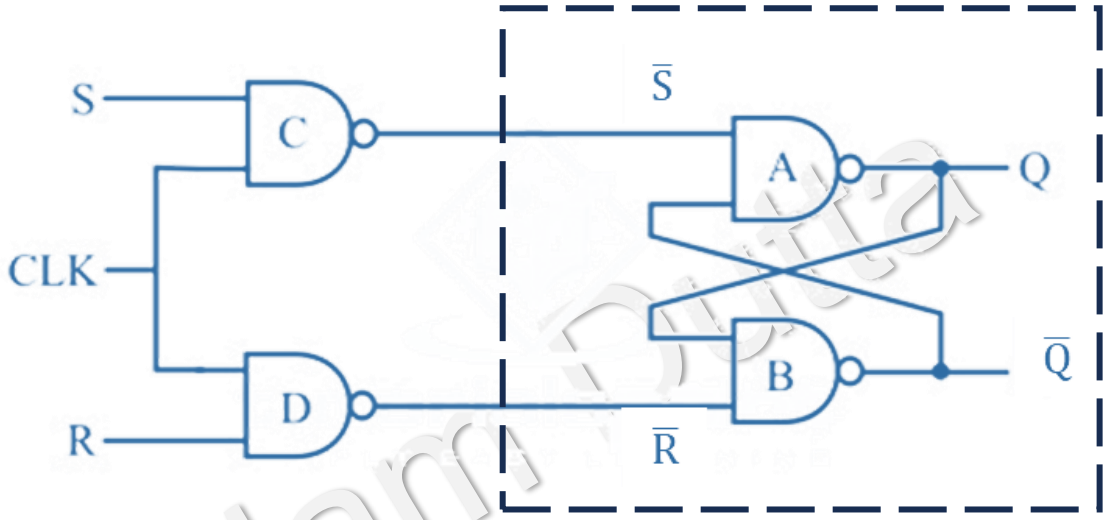


(b) Positive-edge response

Flip-Flops

SR Flip-Flop





NAND SR Latch

Truth Table

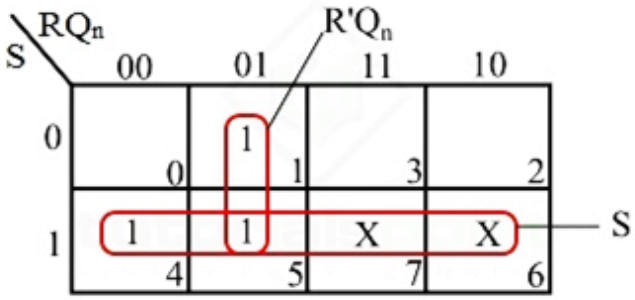
Clk	S	R	Q_{n+1}	State
0	x	x	Q_n	Hold
1	0	0	Q_n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	x	Invalid

Characteristic Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

Excitation Table

Q_n	Q_{n+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0



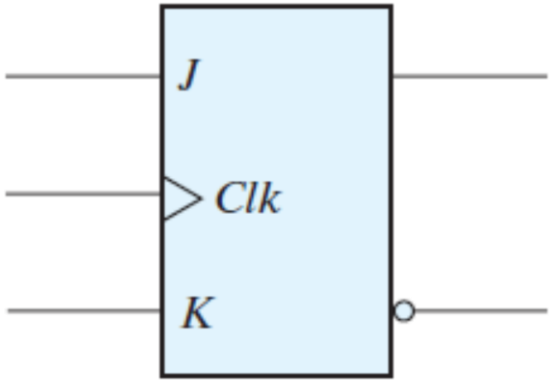
Characteristic Equation:

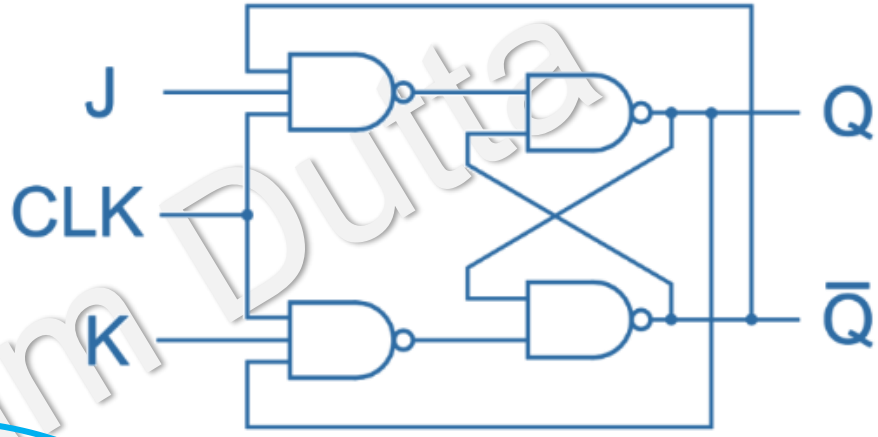
$$Q_{n+1} = S + \overline{R}Q_n$$

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Flip-Flops

JK (Jack-Kilby) Flip-Flop: if both J and K are high, then at clock edge, output toggles from one state to other, thus mitigating the drawback of SR flip-flop.





Truth Table

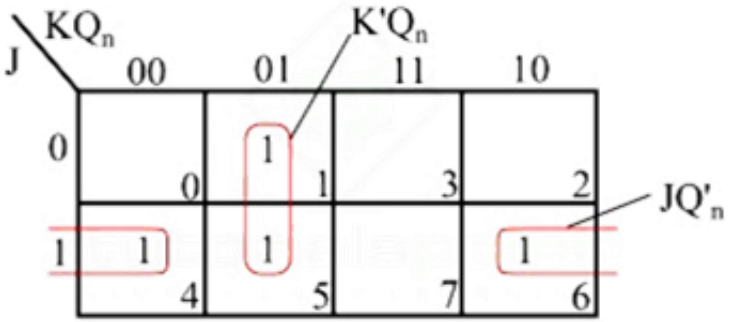
Clk	J	K	Q_{n+1}	State
0	x	x	Q_n	Hold
1	0	0	Q_n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	$\overline{Q_n}$	Toggle

Characteristic Table

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Excitation Table

Q_n	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0



Characteristic Equation:

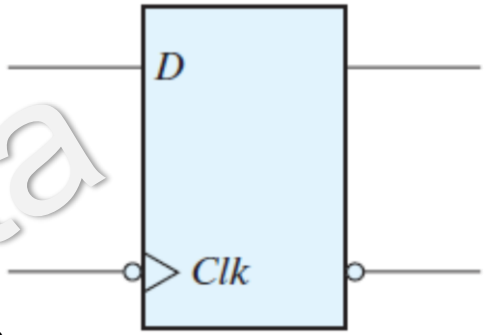
$$Q_{n+1} = J\overline{Q_n} + \overline{K}Q_n$$

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Flip-Flops

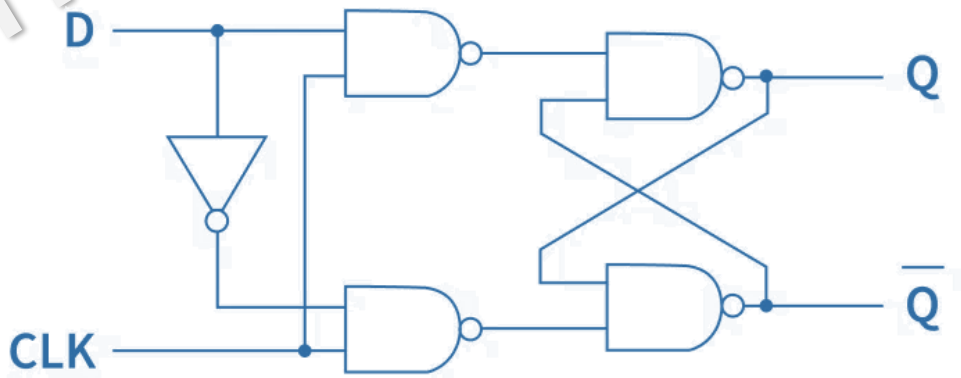
D (Data or Delay) Flip-Flop

- Output (Q) is same as input and changes only at active transition of clock
- If D = 1 and the positive going clock edge appears, Q = 1 and $\overline{Q} = 0$
- When D = 0 and positive going pulse appears, Q = 0 and $\overline{Q} = 1$
- For negative going edge of the incoming clock pulse, the flip-flop is inactive don't care



Truth Table

D	Q_{n+1}	State
0	0	Reset
1	1	Set



Characteristic Table

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Characteristic Equation: $Q_{n+1} = D\overline{Q_n} + DQ_n$

$= D(\overline{Q_n} + Q_n)$

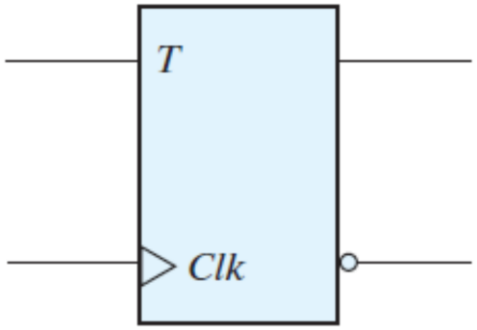
$= D$

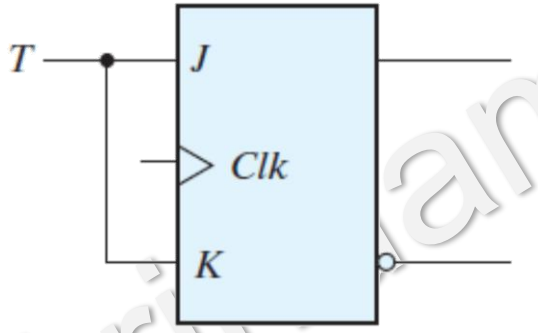
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Flip-Flops

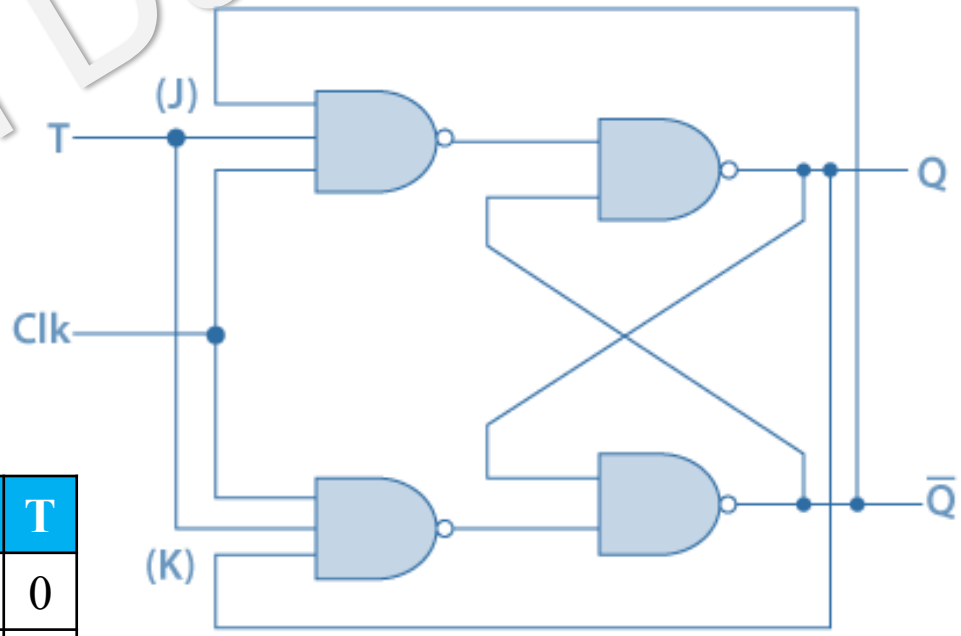
T (Toggle) Flip-Flop

- JK flip-flops with both J and K terminals connected together (shorted) permanently
- If T = 0, J = 0 and K = 0, the output Q and $\overline{Q}$ will remain unchanged
- When T = 1, J = 1 and K = 1, output will toggle corresponding to each and every edge of the clock signal





From JK flip-flop



Truth Table

T	Q_{n+1}	State
0	Q	Hold
1	$\overline{Q}$	Toggle

Characteristic Table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic Equation:

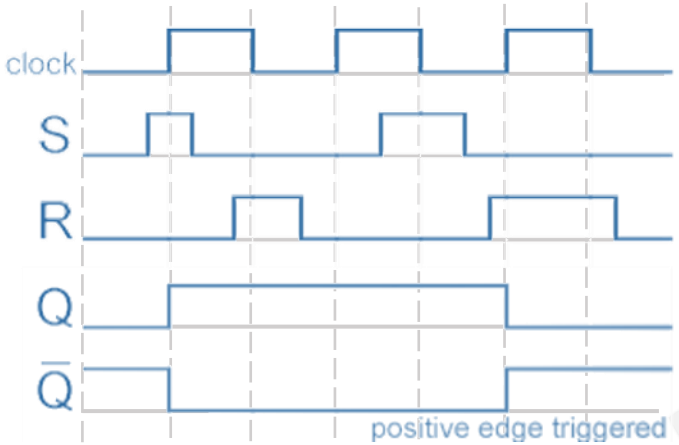
$$Q_{n+1} = T\overline{Q_n} + \overline{T}Q_n = T \oplus Q_n$$

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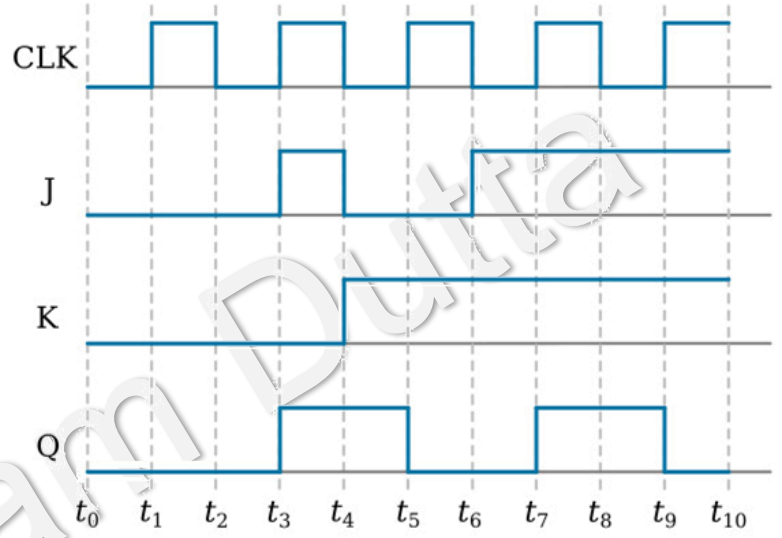
Flip-Flops

Timing Diagrams of Flip-Flops

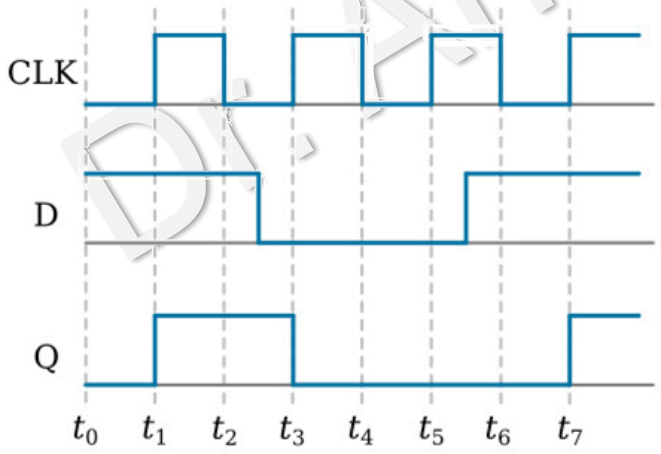
SR flip-flop



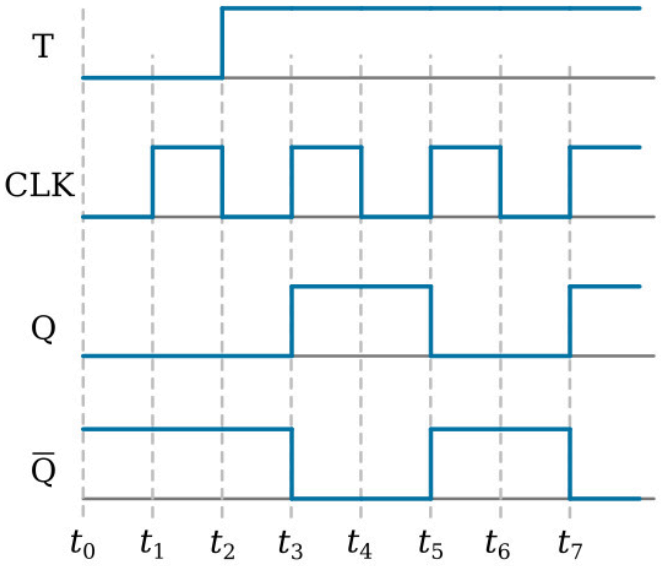
JK flip-flop



D flip-flop



T flip-flop



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Latches and Flip-Flops

Latch	Flip-flop
A latch is an electronic sequential logic circuit used to store information in an asynchronous arrangement.	A flip-flop is an electronic sequential logic circuit used to store information in an synchronous arrangement. It has two stable states and maintains its states for an indefinite period until a trigger pulse is applied.
One latch can store one bit information, but output state changes only in response to data input.	One flip-flop can store one bit data, but output state changes with trigger pulse only.
Latch is an asynchronous device and it has no clock input.	Flip-flop has clock input and its output is synchronised with clock pulse.
Latch holds a bit value and it remains constant until new inputs force it to change.	Flip-flop holds a bit value and it remains constant until a trigger pulse is received.
Latches are level-sensitive and the output tracks the input when the level is high. Therefore as long as the level is logic level 1, the output can change if the input changes.	Flip-flops are edge-sensitive. They can store the input only when there is either a rising or falling edge of the clock.

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Problem

Example: Determine the output waveform of a NAND S-R latch when the inputs as shown in the first figure are applied. Consider initial output of S-R Latch is low and enable is always high.

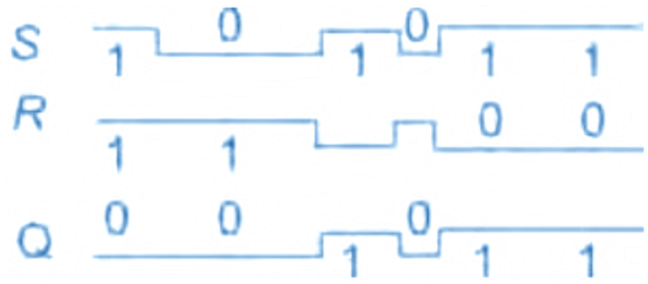


The output waveform of S-R Latch is illustrated below.

Initial output of S-R latch Q is 0.

When S=1, R=1, this is the invalid state of SR Latch and output Q is 0. After that S=0 and R=1, the latch is reset. So, the output Q has not changed and it retain at 0.

Then S=1 R=0, latch is set. So, the output Q changes from 0 to 1. Similarly, other outputs of SR latch can be justified.



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Problem

Example: Write the output of a clocked *S-R* flip-flop when the inputs are *R* = 011001 and *S* = 101100. Assume initially *Q* is low.

The output of *S-R* flip-flop Q_{n+1} is shown in the Table.

Initial output Q_n is 0. If $S = 1$, $R = 0$, $Q_n = 0$ and first clock pulse is applied, the next state output Q_{n+1} will be 1.

When $S = 0$, $R = 1$, $Q_n = 1$ and second clock pulse is applied, the next state output will be 0.

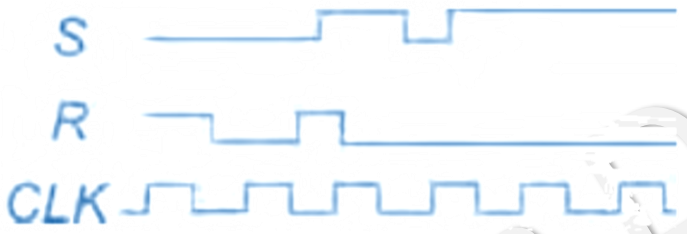
In the same way, remaining next state outputs with respect to clock pulse can be justified using truth table of *S-R* flip-flop.

Inputs			Outputs	
CLK	*S*	*R*	Q_n	Q_{n+1}
↑	1	0	0	1
↑	0	1	1	0
↑	1	1	0	0
↑	1	0	0	1
↑	0	0	1	1
↑	0	1	1	0

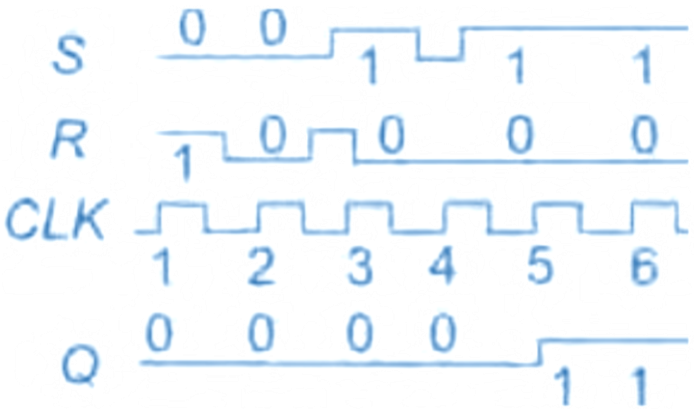
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Problem

Example: Draw the output waveform of a positive edge triggered S-R flip-flop when the inputs are given in the first figure. Assume initially flip-flop output is low.



If S = 1, R = 0 and positive edge triggers applied, the flip-flop will be set. If S = 0, R = 1 flip-flop will be reset when positive edge triggered applied.



At the instant of first clock pulse, S = 0 and R = 1, flip-flop will be reset and output Q is 0.

During the second clock pulse, S = 0 and R = 0 and flip-flop should retain its previous state and there will be no change in output. So, the output Q = 0.

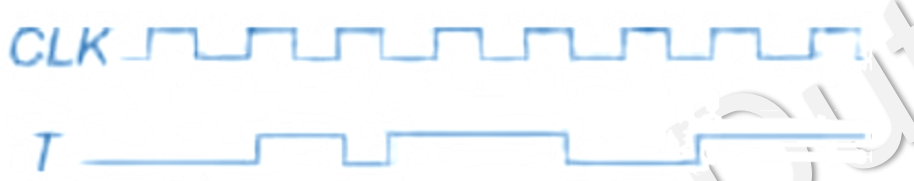
Subsequently, S = 1 and R = 1, flip-flop operates in invalid state and output Q = 0.

At the instant of fourth clock pulse, S = 0 and R = 0, the output of the flip-flop Q = 0. During the fifth clock pulse, S = 1 and R = 0, flip-flop is set and output Q = 1.

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Problem

Example: The waveforms as depicted in the figure are applied to a T flip-flop. Draw the output waveform of the clocked T flip-flop.

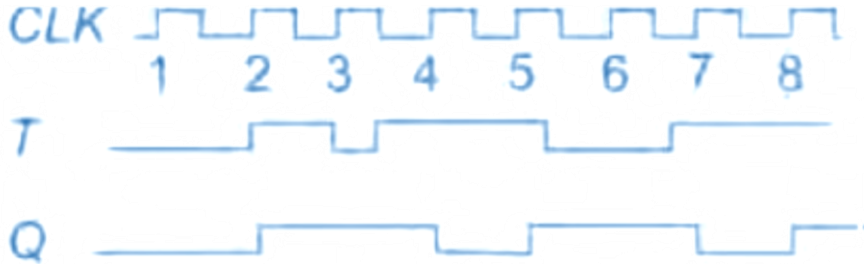


Assuming a positive edge-triggered T flip flop and the initial state of output Q to be 0:

During the first clock pulse T = 0, then output Q will be 0.

At the instant of second clock pulse, T=1, the output will be the complement of previous state. So, the output Q is 1.

Similarly, we can determine the output of T flip-flop for remaining clock signal and T input.



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Problem

Example: A positive-edge-triggered JK flip-flop is initially in the RESET state ($Q_0 = 0$). The following sequence of J and K inputs is applied at successive rising edges of the clock:

Clk	J	K
1	1	0
2	1	1
3	0	1
4	1	1
5	0	0
6	1	1
7	1	0
8	0	1

Determine (a) Q after each clock edge; (b) complete sequence of states of the flip-flop; (c) clock edges at which the flip-flop toggles; (d) final state of the flip-flop after 8th clock edge.

(a), (b)

Clk	J	K	Q_{n+1}	State
Initial	-	-	0	-
1	1	0	1	Set
2	1	1	0	Toggle
3	0	1	0	Reset
4	1	1	1	Toggle
5	0	0	1	Hold
6	1	1	0	Toggle
7	1	0	1	Set
8	0	1	0	Reset

Sequence: 0, 1, 0, 0, 1, 1, 0, 1, 0

(c) The flip-flop toggles at 2nd, 4th, and 6th clock edges.

(d) $Q_8 = 0$ (Reset)

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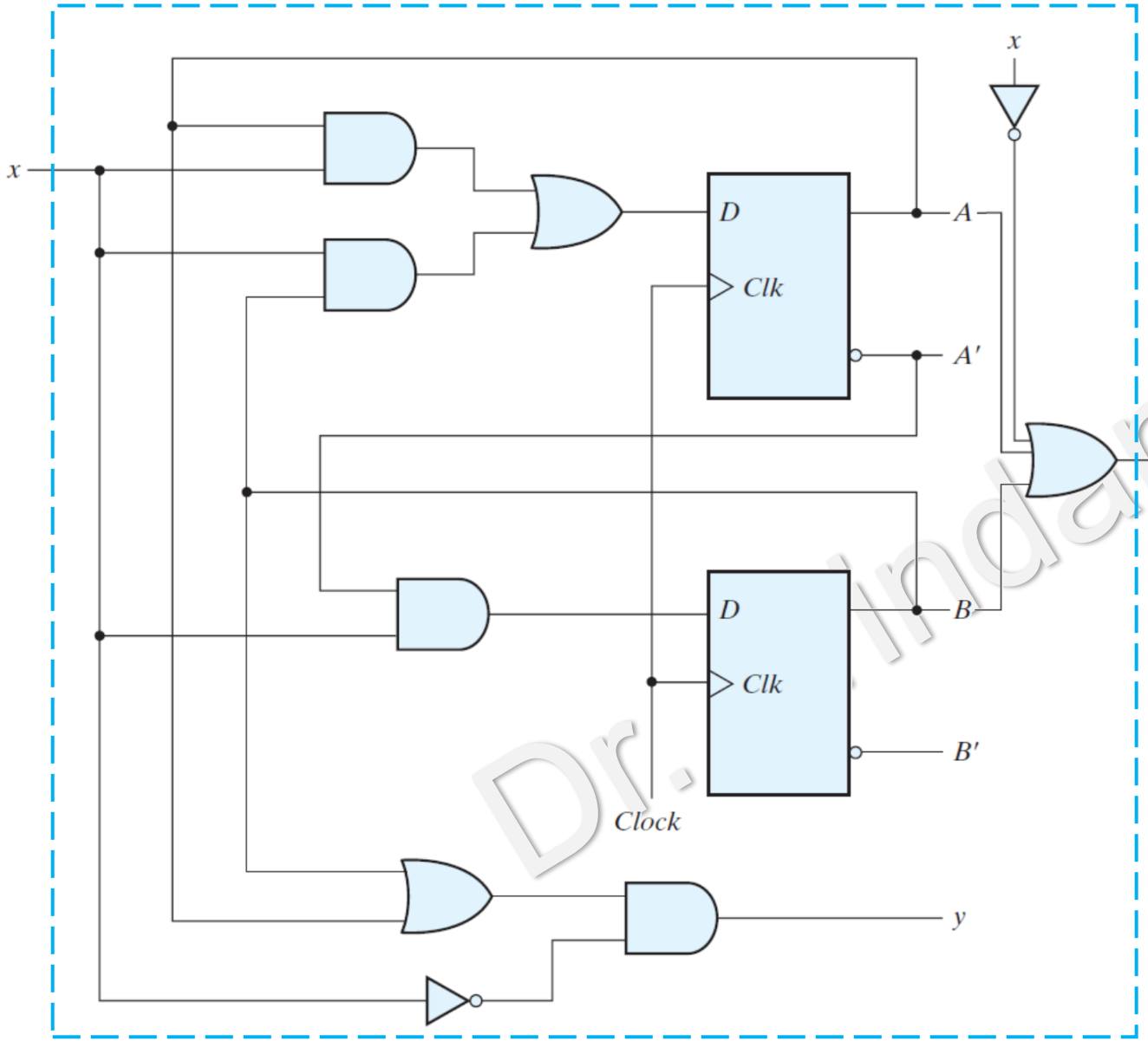
ECE2002
DIGITAL LOGIC DESIGN

Module 3 – Lecture 2

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State Table



Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

- A state (or transition) table enumerates the time sequence of inputs, outputs, and flip-flop states.
- A sequential circuit with m flipflops and n inputs needs 2^{m+n} rows in the state table. The binary numbers from 0 through 2^{m+n} - 1 are listed under the present-state and input columns. The next-state section has m columns, one for each flip-flop. The output section has as many columns as there are output variables.

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State Table

- The state table can be expressed in a slightly different form having only three sections: present state, next state, and output.
- The input conditions are enumerated under the next-state and output sections.
- For each present state, there are two possible next states and outputs, depending on the value of the input.
- One form may be preferable to the other, depending on the application.

Present State		Input	Next State		Output
A	*B*	*x*	*A*	*B*	*y*
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

OR

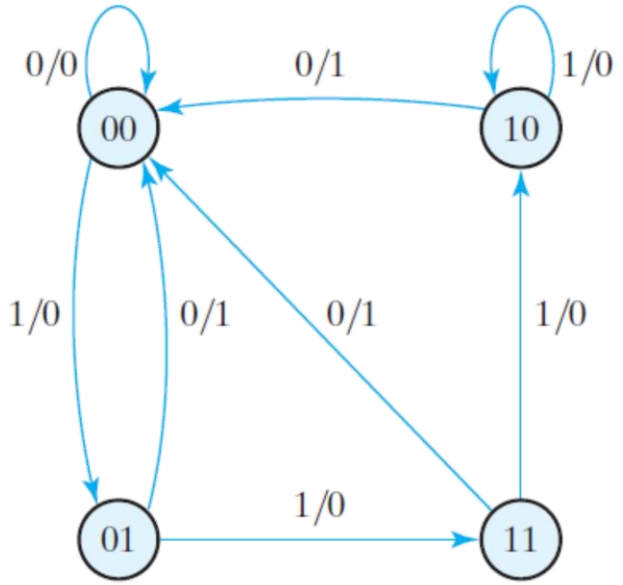
Present State		Next State				Output	
		x = 0		*x* = 1		*x* = 0	*x* = 1
A	*B*	*A*	*B*	*A*	*B*	*y*	*y*
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

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State Diagram

- Graphical representation of information in state table.
- A state is represented by a circle, and the transitions between states are indicated by directed lines connecting the circles.
- For a circuit with two flip-flops, there will be $2^2 = 4$ states: $S_0 \rightarrow 00$, $S_1 \rightarrow 01$, $S_2 \rightarrow 10$, and $S_3 \rightarrow 11$.
- The binary number inside each circle identifies the state of the flip-flops.
- The directed lines are labeled with two binary numbers separated by a slash. The input value during the present state is labeled first, and the number after the slash gives the output during the present state with the given input.
- A directed line connecting a circle with itself indicates that no change of state occurs.

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0



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State Diagram

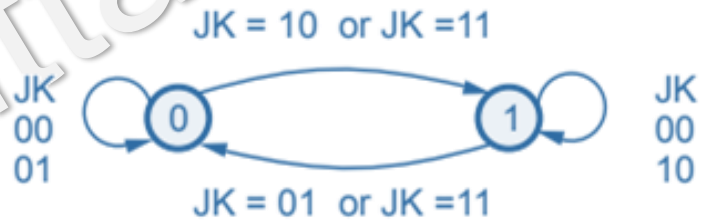
Excitation Tables and State Diagrams of Flip-Flops

Q_n	Q_{n+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0



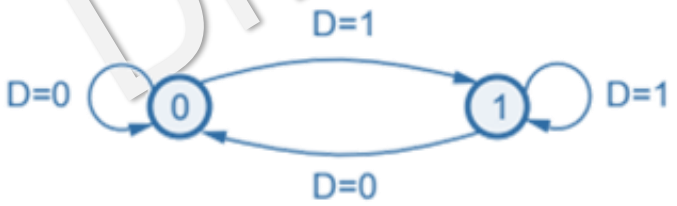
SR flip-flop

Q_n	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0



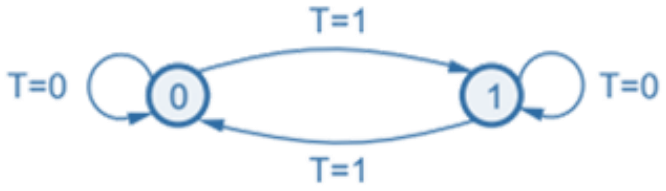
JK flip-flop

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1



D flip-flop

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



T flip-flop

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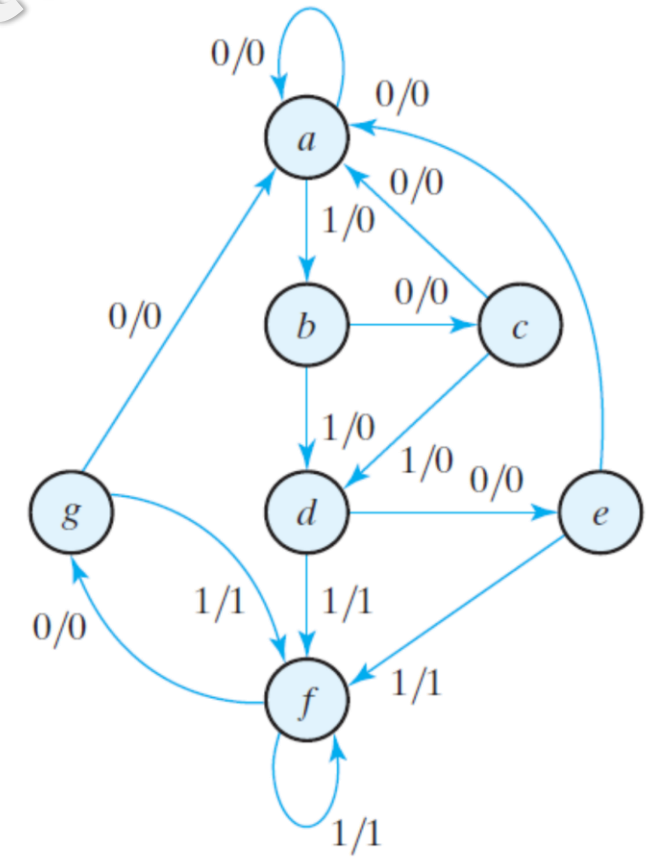
State Reduction

- Reduction in the number of flip-flops in a sequential circuit.
- Involves reducing the number of states in a state table, while keeping the external input–output requirements unchanged.

Example: The state diagram on the right represents some sequential circuit.

From the state diagram, we obtain the output and state sequence for the given input sequence and draw the state table.

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1



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State Reduction

- Two states are said to be equivalent if, for each member of the set of inputs, they give exactly the same output and send the circuit either to the same state or to an equivalent state.
- When two states are equivalent, one of them can be removed without altering the input–output relationships.
- States *e* and *g* both go to states *a* and *f* and have outputs of 0 and 1 for $x = 0$ and $x = 1$, respectively. Therefore, states *g* and *e* are equivalent, and one of these states can be removed.
- The row with present state *g* is removed, and state *g* is replaced by state *e* each time it occurs in the columns headed "Next State."

	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*g*	*f*	0	1
g	*a*	*f*	0	1



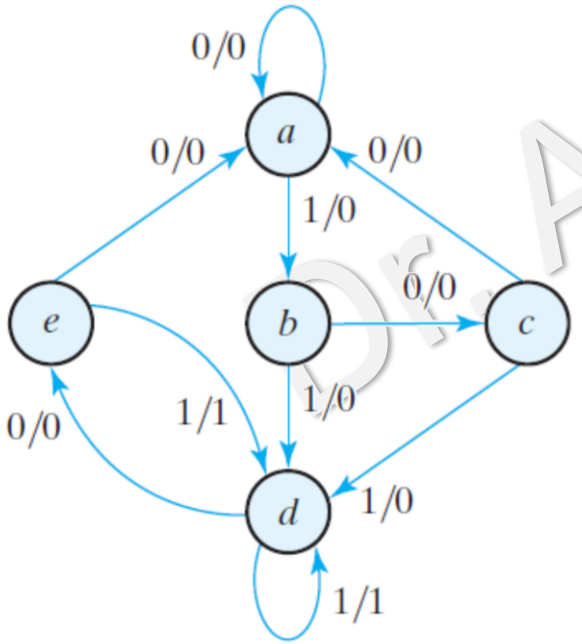
	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*e*	*f*	0	1

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State Reduction

- Present state *f* now has next states *e* and *f* and outputs 0 and 1 for $x = 0$ and $x = 1$, just like present state *d*.
- Therefore, states *f* and *d* are equivalent, and state *f* can be removed and replaced by *d*.
- Now, the state diagram for the reduced table would consist of only five states.

Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*e*	*f*	0	1



Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*d*	0	1
e	*a*	*d*	0	1

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State Assignment

- In order to design a sequential circuit with physical components, it is necessary to assign unique coded binary values to the states.
- For a circuit with m states, the codes must contain n bits, where $2^n \geq m$.
- For example, with three bits, it is possible to assign codes to eight states, denoted by binary numbers 000 through 111.
- If the unreduced state table of the previous example is used, we must assign binary values to seven states; the remaining state is unused.
- If the reduced state table is used, only five states need binary assignment, and we are left with three unused states.
- Unused states are treated as don't-care conditions during the design.

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State Assignment

- The simplest way to code five states is to use the first five integers in binary counting order.
- Another similar assignment is the Gray code. Here, only one bit in the code group changes when going from one number to the next.
- Another possible assignment often used is the one-hot assignment. This configuration uses as many bits as there are states in the circuit. At any given time, only one bit is equal to 1 while all others are kept at 0.
- The second table is the reduced state table of the previous example with binary assignment 1 substituted for the letter symbols of the states.
- A different assignment will result in a state table with different binary values for the states.

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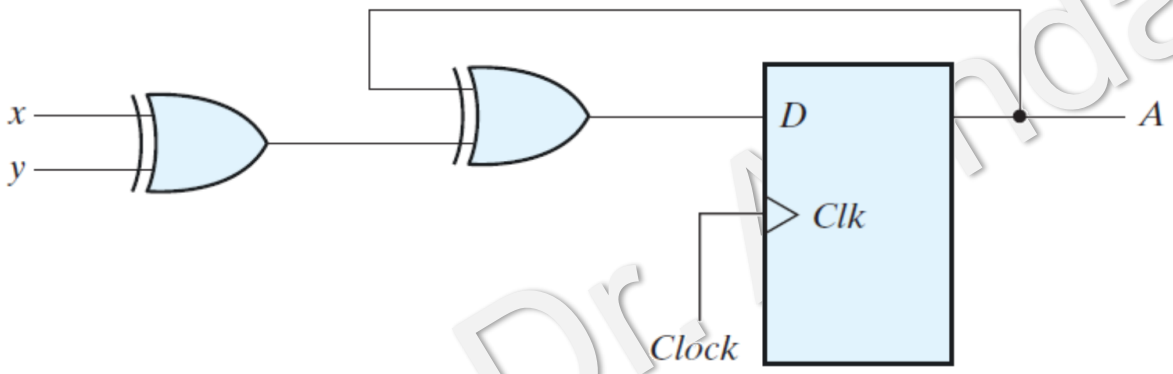
State	Assignment 1, Binary	Assignment 2, Gray Code	Assignment 3, One-Hot
a	000	000	00001
b	001	001	00010
c	010	011	00100
d	011	010	01000
e	100	110	10000

	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
000	000	001	0	0
001	010	011	0	0
010	000	011	0	0
011	100	011	0	1
100	000	011	0	1

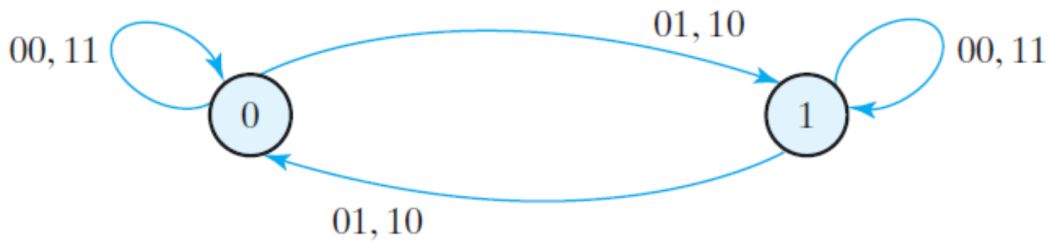
Analysis of Sequential Circuits Using Flip-Flops

Using D Flip-Flop

- Suppose, the circuit we want to analyze is described by the input equation $D_A = A \oplus x \oplus y$
- The D_A symbol implies a D flip-flop with output A. Variables x and y are the inputs to the circuit. No output equations are given, which implies that the output comes from the output of the flip-flop. We draw the circuit diagram.
- The next-state values are obtained from the state equation $A(t+1) = A \oplus x \oplus y$



Present state	Inputs		Next state
A	x	y	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



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Analysis of Sequential Circuits Using Flip-Flops

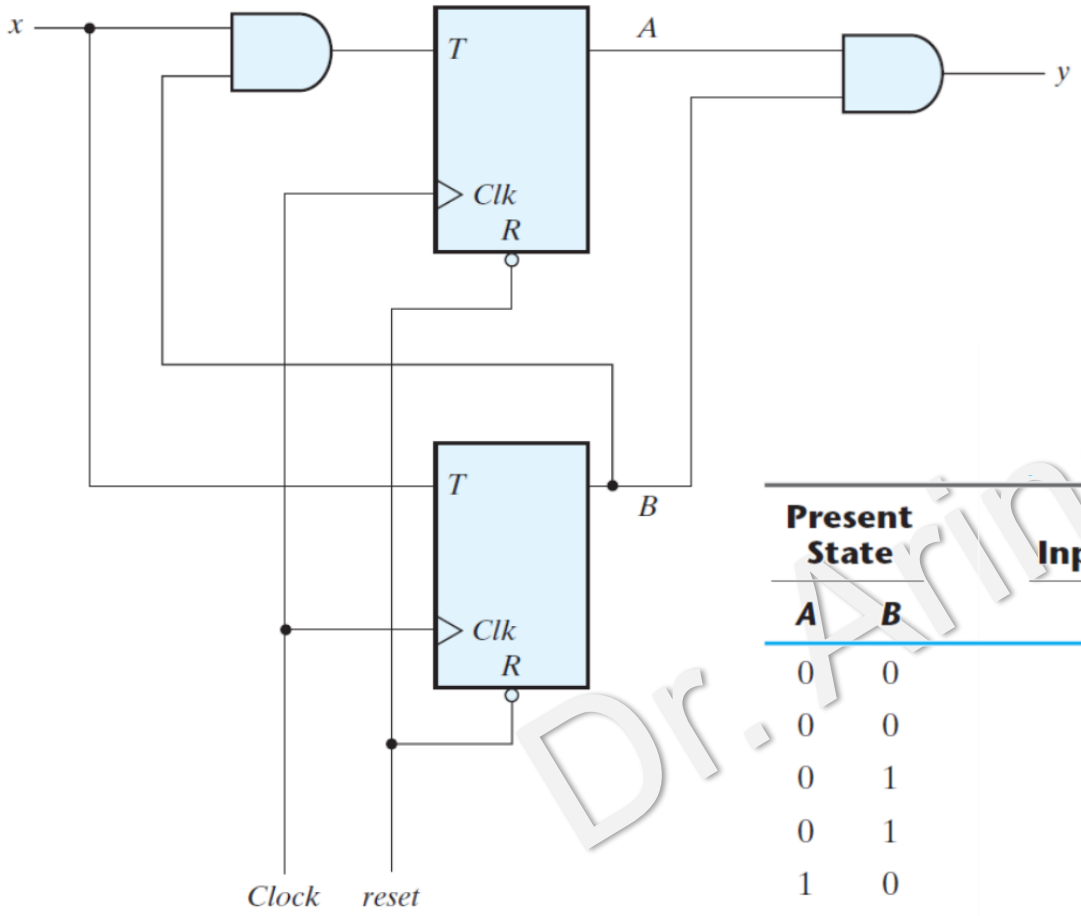
Using T Flip-Flop

- For a *D*-type flip-flop, the state equation is the same as the input equation.
- To determine the next-state values of a sequential circuit that uses T -type flip-flops:
  - Determine the flip-flop input equations in terms of the present state and input variables.
  - Substitute the input equations into the flip-flop characteristic equation to obtain the state equations.
  - Use the corresponding state equations to determine the next-state values in the state table.

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Analysis of Sequential Circuits Using Flip-Flops

Using T Flip-Flop



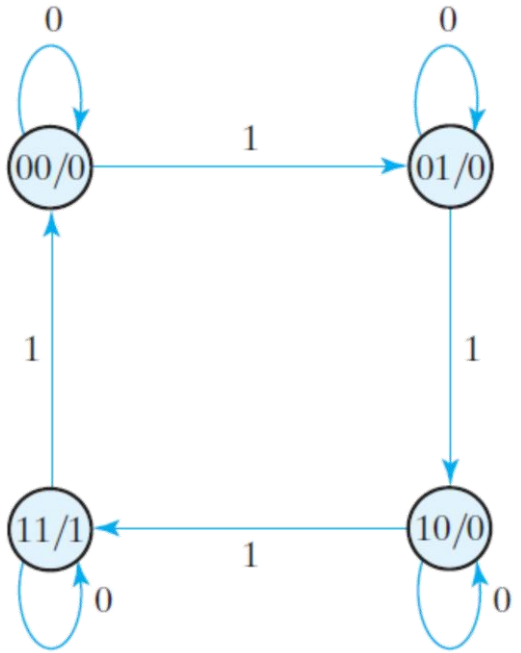
From the given circuit, the input and output equations are:

$$T_A = Bx;\ T_B = x;\ y = AB$$

The values for the next state can be derived from the state equations by substituting T_A and T_B in the characteristic equation of T flip-flop ($Q_{n+1} = T \oplus Q_n = T\overline{Q_n} + \bar{T}Q_n$), yielding

$$A(t+1) = (Bx)'A + (Bx)A' = AB' + Ax' + A'Bx$$
$$B(t+1) = x \oplus B$$

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1



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Module 3 – Lecture 3

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Design Procedure of Sequential Circuits

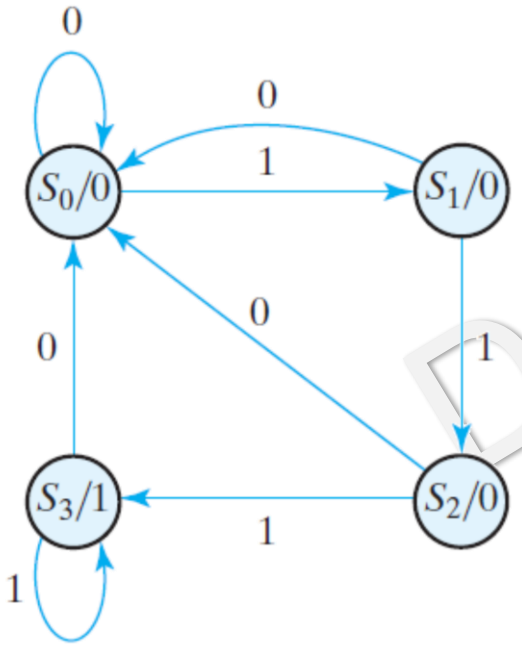
- The procedure for designing synchronous sequential circuits involves the following steps:
  - Derive a state diagram for the circuit based on the given specifications.
  - Reduce the number of states if necessary.
  - Assign binary values to the states.
  - Obtain the binary-coded state table.
  - Choose the type of flip-flops to be used.
  - Derive the simplified flip-flop input equations and output equations.
  - Draw the logic diagram.

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Design Procedure of Sequential Circuits

Example: We wish to design a circuit that detects a sequence of three or more consecutive 1's in a string of bits coming through an input line.

The state diagram is derived by starting with state S_0. If the input is 0, the circuit stays in S_0, but if the input is 1, it goes to state S_1 to indicate that a 1 was detected. If the next input is 1, it changes to state S_2 to indicate the arrival of two consecutive 1's, but if the input is 0, the state goes back to S_0. The third consecutive 1 sends the circuit to state S_3. If more 1's are detected, the circuit stays in S_3. Any 0 input sends the circuit back to S_0. In this way, the circuit stays in S_3 as long as there are three or more consecutive 1's received (Moore model sequential circuit).



Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

The state table is derived from the state diagram with a sequential binary assignment.

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Design Procedure of Sequential Circuits

We choose two D flip-flops to represent the four states, and we label their outputs A and B. There is one input x and one output y. The characteristic equation of the D flip-flop is $Q(t + 1) = D_Q$, which means that the next-state values in the state table specify the D input condition for the flip-flop. The flip-flop input equations can be obtained directly from the next-state columns of A and B and expressed in sum-of-minterms form as

$$A(t + 1) = D_A(A, B, x) = \Sigma(3, 5, 7) \qquad B(t + 1) = D_B(A, B, x) = \Sigma(1, 5, 7) \qquad y(A, B, x) = \Sigma(6, 7)$$

where A and B are the present-state values of flip-flops A and B, x is the input, and D_A and D_B are the input equations. The minterms for output y are obtained from the output column in the state table. After simplification using K-Maps:

$D_A = Ax + Bx$

A \ Bx	00	01	11	10
0	m_0	m_1	m_3 1	m_2
1	m_4	m_5 1	m_7 1	m_6

x | B

$D_B = Ax + B'x$

A \ Bx	00	01	11	10
0	m_0	m_1 1	m_3	m_2
1	m_4	m_5 1	m_7 1	m_6

x | B

$y = AB$

A \ Bx	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7 1	m_6 1

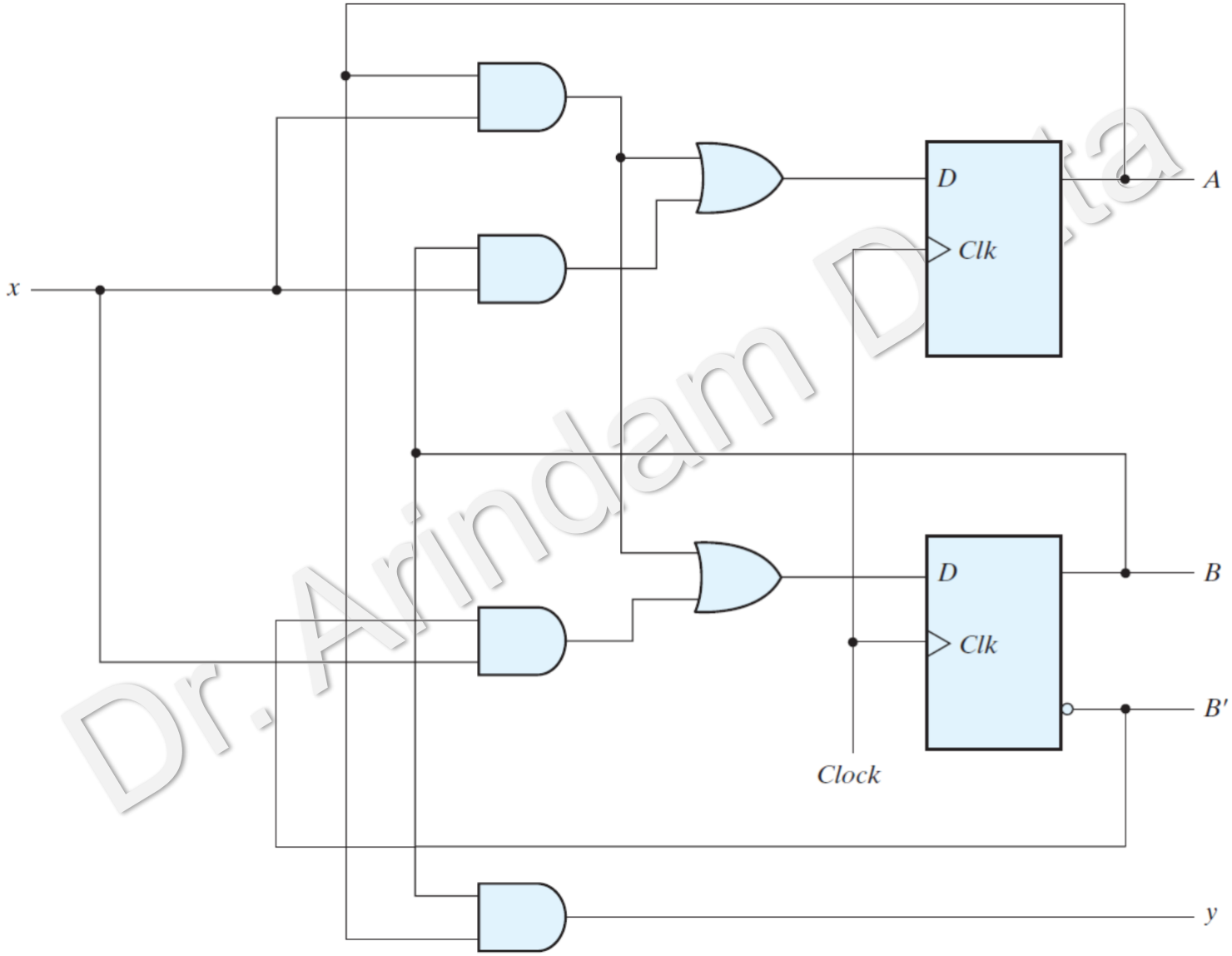
x | B

The advantage of designing with D flip-flops is that the Boolean equations describing the inputs to the flip-flops can be obtained directly from the state table.

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Design Procedure of Sequential Circuits

Final Circuit:

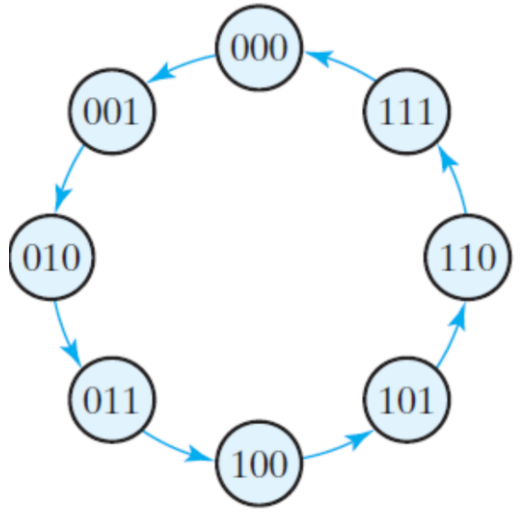


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Design Procedure of Sequential Circuits

Example: An n -bit binary counter consists of n flip-flops that can count in binary from 0 to 2^n - 1. Let us design a 3-bit binary counter. The flip-flop outputs repeat the binary count sequence with a return to 000 after 111.

In the state diagram, the directed lines between circles are not marked with input and output values as the only input to a counter circuit is the clock.



The outputs are specified by the present state of the flip-flops. The next state of a counter depends entirely on its present state, and the state transition occurs every time the clock goes through a transition.

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	T_{A2}	T_{A1}	T_{A0}
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

Binary counters are constructed most efficiently with T flip-flops because of their complement property.

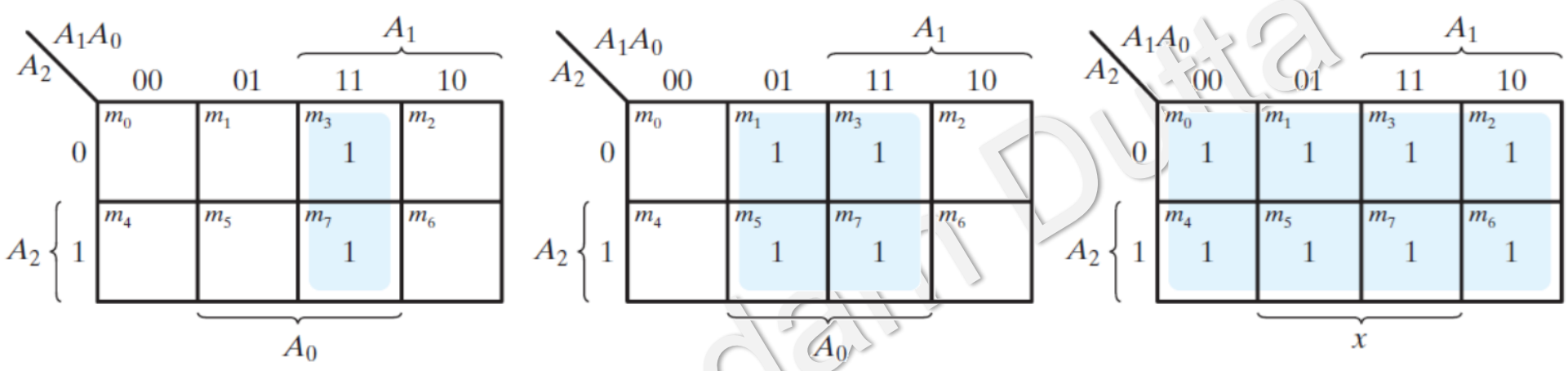
The flip-flop excitation for the T inputs is derived from the excitation table of the T flip-flop and by inspection of the state transition of the present state to the next state.

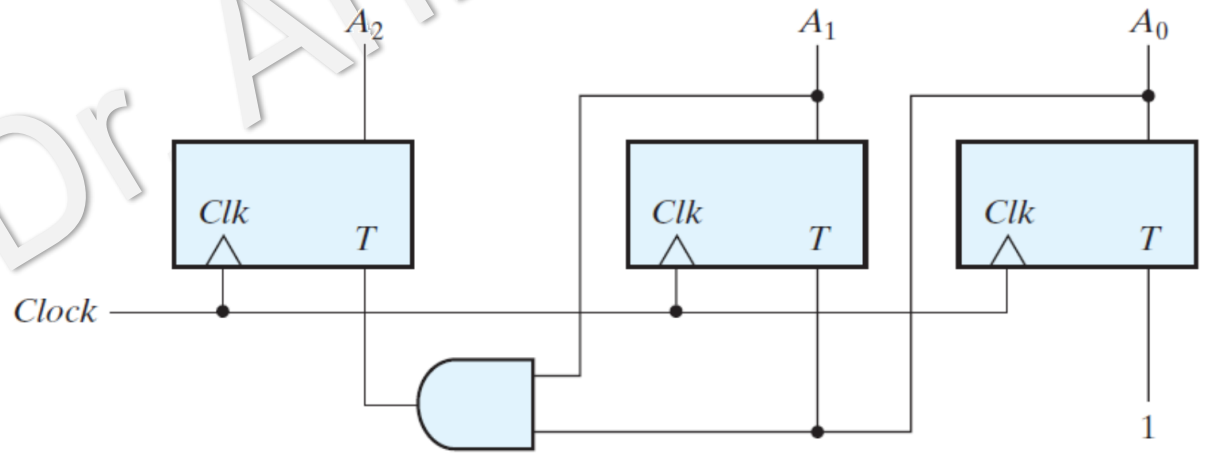
The last row, with present state 111, is compared with the first count 000, which is its next state.

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Design Procedure of Sequential Circuits

The flip-flop input equations are simplified using K-maps.





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Design Procedure of Sequential Circuits

Example: Design a sequential circuit for the given state table using D flip-flops. Assume two inputs are A and B, outputs of the sequential circuit are outputs of D flip-flops, present state = S, Next State = S*.

Present state (Q_1Q_0)	Inputs (AB) 00	01	10	11
00	01	00	00	01
01	10	00	00	10
10	11	00	00	11
11	01	00	00	01
	Next State (Q_1^* Q_0^*)			

Present state (Q_1Q_0)	Inputs (AB) 00	01	10	11
00	01	00	00	01
01	10	00	00	10
10	11	00	00	11
11	01	00	00	01
	Inputs of flip-flops D_1 D_0			

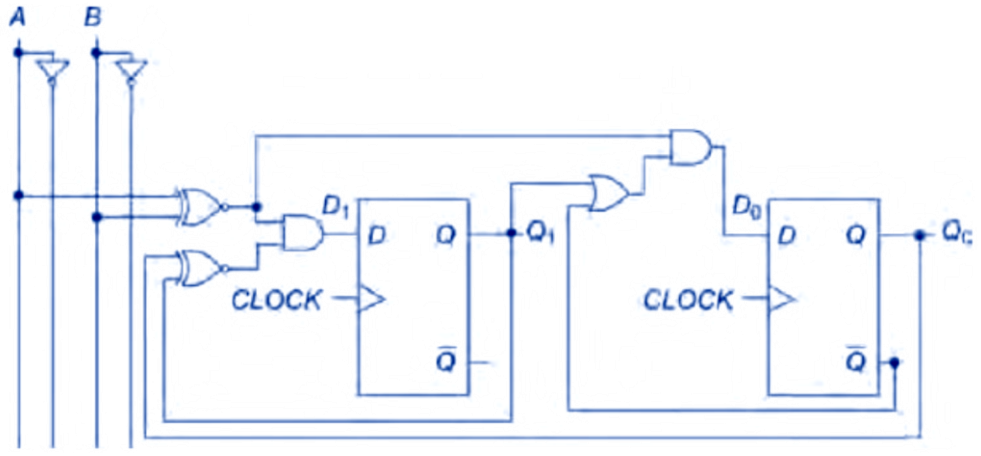
Present state (S)	Inputs (AB) 00	01	10	11
S_0	S_1	S_0	S_0	S_1
S_1	S_2	S_0	S_0	S_2
S_2	S_3	S_0	S_0	S_3
S_3	S_1	S_0	S_0	S_1
	Next State (S^*)			

Q_1Q_0 \ AB	$\bar{A}\bar{B}$ 00	$\bar{A}B$ 01	AB 11	$A\bar{B}$ 10
$\bar{Q}_1\bar{Q}_0$ 00	0	0	0	0
$\bar{Q}_1Q_0$ 01	1	0	1	0
Q_1Q_0 11	0	0	0	0
$Q_1\bar{Q}_0$ 10	1	0	1	0

D_1

Q_1Q_0 \ AB	$\bar{A}\bar{B}$ 00	$\bar{A}B$ 01	AB 11	$A\bar{B}$ 10
$\bar{Q}_1\bar{Q}_0$ 00	1	0	1	0
$\bar{Q}_1Q_0$ 01	0	0	0	0
Q_1Q_0 11	1	0	1	0
$Q_1\bar{Q}_0$ 10	1	0	1	0

D_0



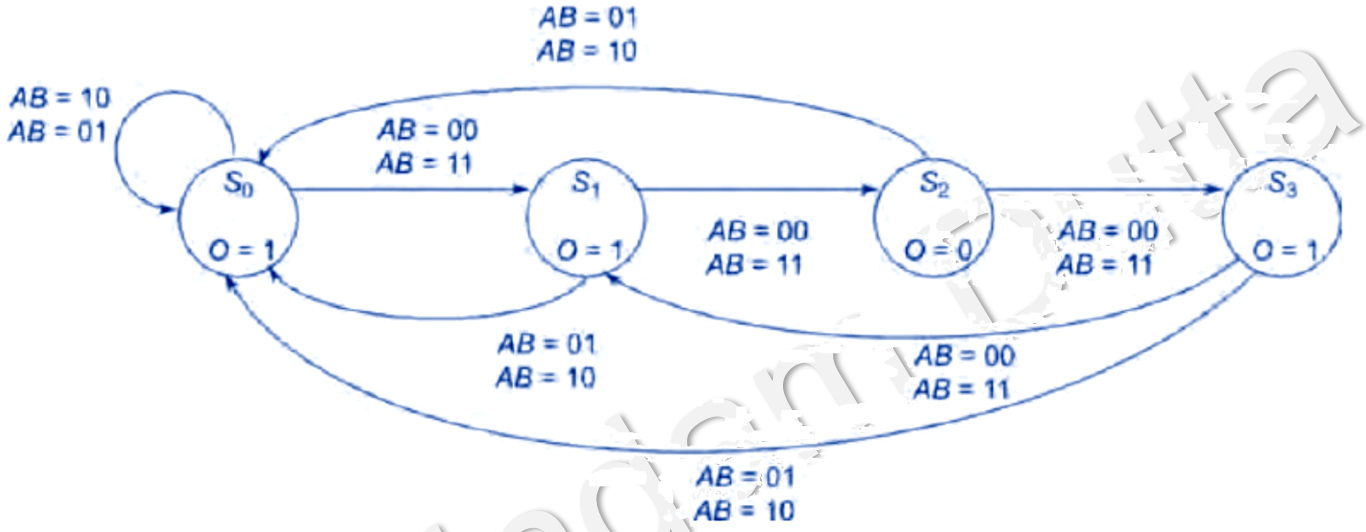
$$D_1 = Q_1^* = Q\bar{Q}_0\bar{A}\bar{B} + \bar{Q}_1Q_0\bar{A}\bar{B} + Q_1\bar{Q}_0AB + \bar{Q}_1Q_0AB = (\overline{A \oplus B})\,(Q_1 \oplus Q_0)$$

$$D_0 = Q_0^* = \bar{A}\bar{B}Q_1 + ABQ_1 + \bar{A}\bar{B}\bar{Q}_0 + AB\bar{Q}_0 = (\overline{A \oplus B})\,(Q_1 + \bar{Q}_0)$$

Dr. Indrajit Dutta

Design Procedure of Sequential Circuits

Example: The state diagram of a sequential circuit is given. Draw the state table. The two inputs are A and B, output is O.



Present state (S)	Inputs (AB) 00	01	10	11
S_0	S_1	S_0	S_0	S_1
S_1	S_2	S_0	S_0	S_2
S_2	S_3	S_0	S_0	S_3
S_3	S_1	S_0	S_0	S_1
	Next State (S^*)			

Present state (Q_1Q_0)	Inputs (AB) 00	01	10	11
00	01/1	00/1	00/1	01/1
01	10/1	00/1	00/1	10/1
10	11/0	00/0	00/0	11/0
11	01/1	00/1	00/1	01/1
	Next State (Q_1^* Q_0^*)/Output(O)			

Dr. Arinaga Datta

Design Procedure of Sequential Circuits

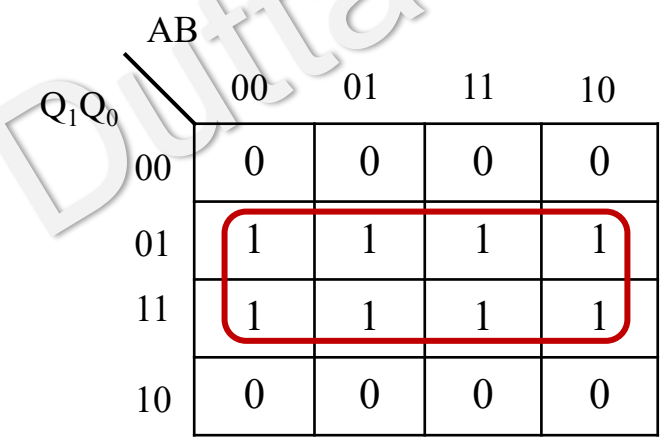
Example: Design a sequential circuit for the given state table using D flip-flops. Assume two inputs are A and B, output of the sequential circuit is O, present state of D flip-flops = Q_1 Q_0, Next State of D flip-flops = (Q_1* Q_0*).

Present state(Q_1Q_0)	Inputs (AB) 00	01	10	11
00	01/0	00/0	00/0	01/0
01	10/1	00/1	00/1	10/1
10	11/0	00/0	00/0	11/0
11	01/1	00/1	00/1	01/1

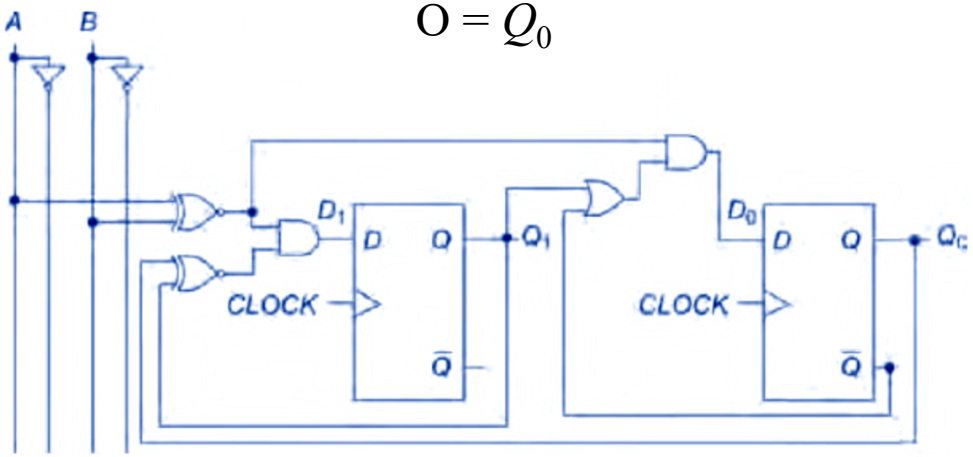
Next State (Q_1* Q_0*)/Output(O)

Present state (Q_1Q_0)	Inputs (AB) 00	01	10	11
00	01/0	00/0	00/0	01/0
01	10/1	00/1	00/1	10/1
10	11/0	00/0	00/0	11/0
11	01/1	00/1	00/1	01/1

Inputs of flip-flops (D_1 D_0)/ Output(O)



$O = Q_0$



From the K-Maps in Slide 8:

$$D_1 = Q_1^* = Q\bar{Q}_0\bar{A}\bar{B} + \bar{Q}_1Q_0\bar{A}\bar{B} + Q_1\bar{Q}_0AB + \bar{Q}_1Q_0AB = (\overline{A \oplus B})\,(Q_1 \oplus Q_0)$$

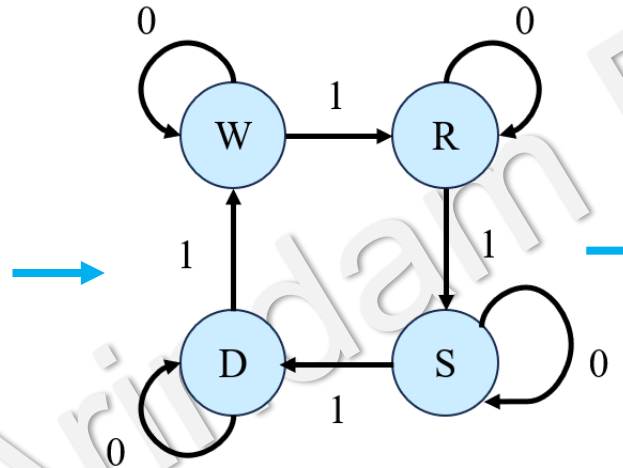
$$D_0 = Q_0^* = \bar{A}\bar{B}Q_1 + ABQ_1 + \bar{A}\bar{B}\bar{Q}_0 + AB\bar{Q}_0 = (\overline{A \oplus B})\,(Q_1 + \bar{Q}_0)$$

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Design Procedure of Sequential Circuits

Example: A four-mode washing machine controller is operated by an external switch. Each time the switch is pressed, the controller changes its mode in the following sequence: Wash → Rinse → Spin → Dry → Wash, and the sequence repeats. Draw the state table and state diagram for this state machine. Assign suitable binary codes to the states and implement the circuit using D flip-flops and logic gates, as required. State any assumptions made.

Present state	Next state	
	Input (x) = 0	Input (x) = 1
W	W	R
R	R	S
S	S	D
D	D	W



Present state (A_nB_n)	Next state ($A_{n+1}B_{n+1}$)	
	x = 0	x = 1
00	00	01
01	01	10
10	10	11
11	11	00

Present state (A_nB_n)	D flip-flop input (D_A D_B)	
	x = 0	x = 1
00	00	01
01	01	10
10	10	11
11	11	00

A_n \ B_n x	00	01	11	10
0	0	1	0	1
1	0	1	0	1

$D_B = B_n' x + B_n x' = B_n \oplus x$

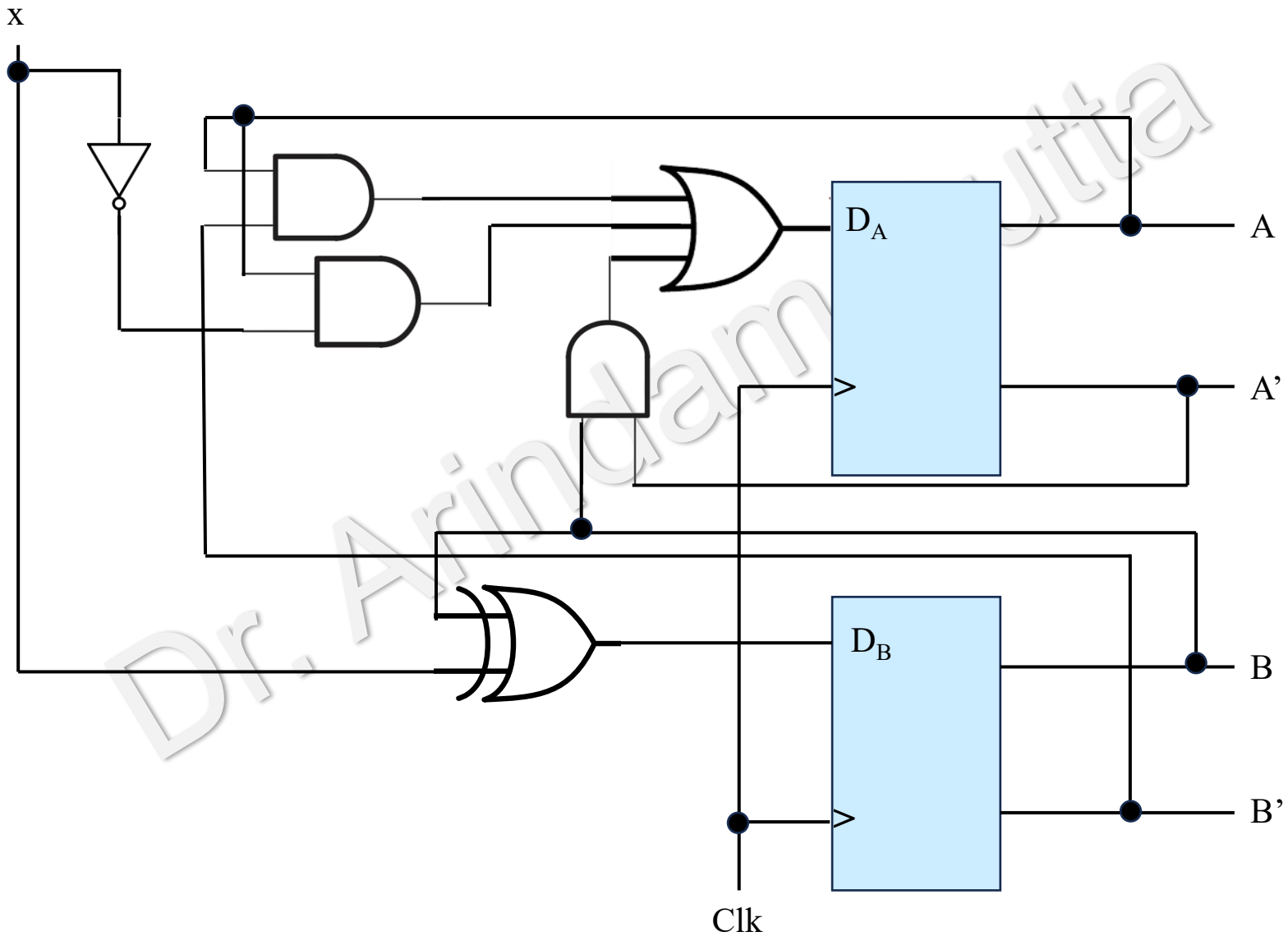
A_n \ B_n x	00	01	11	10
0	0	0	1	0
1	1	1	0	1

$D_A = A_nB_n' + A_n x' + A_n' B_n x$

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Design Procedure of Sequential Circuits

Final Circuit:



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